Nikita Bhansali

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Education

Vellore Institute of Technology — September 2023 – October 2027
Bachelor of Technology in Electronics and Communication Engineering; CGPA:8.48

Skill Set

Embedded Systems & Hardware: ESP32/ESP32-CAM, Arduino IDE, Sensor Interfacing (DHT11, PIR)
RF & Signal Processing: Software Defined Radio (RTL-SDR, HackRF, ADALM-PLUTO), DragonOS
Programming Languages: Python, Embedded C, Java
Databases: MySQL, MongoDB
Frameworks: MERN Stack,GitHub

Internship

MPOnline (Government of MP) | *AI/ML & Advanced Software Engineering Internship* — May 2026 – July 2026
-Built ML models (regression, classification, decision trees, random forest, SVM, clustering, PCA) using Python, NumPy, Pandas, and Scikit-learn, and developed neural networks with TensorFlow/PyTorch for computer vision and NLP.
-Practiced full-stack SDLC: DBMS, SQL, OOP, Git/GitHub, and CRUD app development, followed by requirement analysis, system design, backend development, testing, and cloud deployment for a capstone project.

Projects

Anti-Drone System using SDR | *SDR, RF Systems, Embedded Design* — August 2026 – March 2027
-Engineered a dual-drone counter-UAV system in which a programmable ESP32 "Enemy Drone" simulates rogue RF intrusions across the 2.4 GHz/5.8 GHz ISM bands for campus perimeter security testing.
-Configured a ground station with an ADALM-PLUTO SDR on DragonOS (GNU Radio, SigDigger) to isolate the enemy drone's RF signature and compute real-time signal bearing and RSSI.
-Established a sub-5ms ESP-NOW ground-to-air relay to guide a Defender ESP32 drone toward the target and trigger localized non-kinetic disruption (Wi-Fi de-authentication) to force a controlled landing.

Home Automation with LumaSync | *IoT, Embedded Systems, ESP32* — January 2025 – April 2025
-IoT-based home automation and security system regulating room temperature, detecting motion, and capturing images of entrants, controllable via a local web dashboard and Google Assistant.
-Assembled the system using an ESP32/ESP32-CAM, a DHT11 sensor, a PIR sensor, and a 4-channel relay module; developed a real-time dashboard (HTML/CSS/JS) and integrated Google Assistant via IFTTT.
-Devised predictive, trend-based temperature control logic (dynamic hysteresis) to reduce overshoot and improve system responsiveness.

Smart Surveillance for Public Transport | *Computer Vision, Embedded AI* — September 2024 – January 2025
-Real-time monitoring system for public transport to detect anomalies and enhance passenger safety.
-Implemented OpenCV and YOLO for real-time object detection and video analysis on embedded/edge hardware.
-Fine-tuned anomaly-detection models and optimized YOLO inference for real-time performance on live feeds.

Certifications

Health Hackathon 2026 – Finalist — tinyurl.com/HealthHack26

Health Hackathon 2025 – Finalist — tinyurl.com/HealthHack25

Extracurricular

Languages: Proficient in English, Hindi; learning Japanese and German (250+ day Duolingo streak). — tinyurl.com/DuolingoStreak

Leadership: Student Partner at IEEE and VITronix Club, organizing technical workshops for 1000+ students.

National Service Scheme: Served as Hospitality Lead for the NSS Unit Camp and led the Tree Plantation Drive team.

Hobbies

Anchoring, Dancing, Event Management, Photography, Public Speaking.